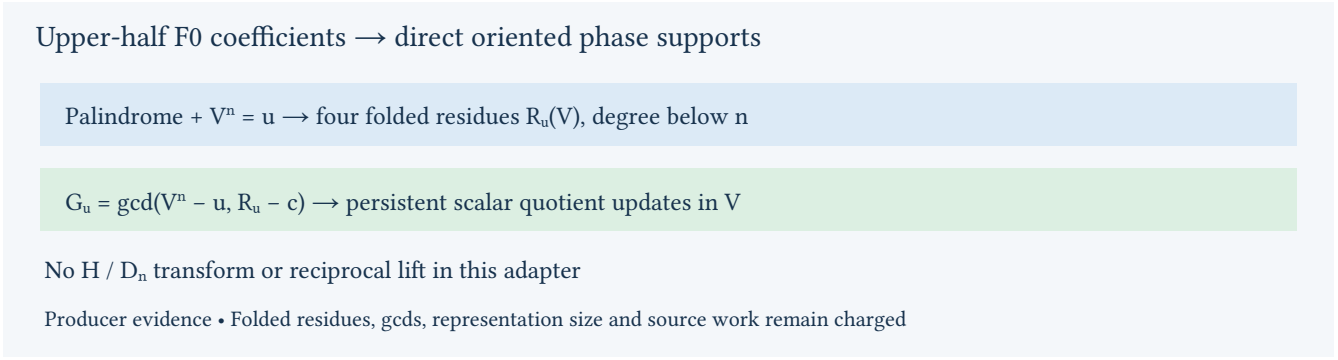
Direct oriented phase supports

T-1877 · 2026-11-07 · Prime-field compressed recovery

The new adapter folds F0 directly modulo each sparse phase equation $V^n = u$. It keeps the resulting oriented support in V and applies later constraints in scalar quotient rings. It does not build the reciprocal H / D_n transform or lift back from W.



Construction. Palindromic symmetry supplies the lower half from the same n+1 upper coefficient deliveries. Each phase gets a residue of degree below n. Intersect with $V^n - u$ using modular exponentiation and gcd; expand the phase polynomial only when the folded equation is identically zero. Normalize back from the fixed anchor V = t0 Z at materialization.

Native control	Observed
Configurations / folded-residue equalities	23 / 92
Oriented supports equal to dense and reciprocal references	13,980
Zero / nonzero-constant equation controls	11 / 376
Initial degree-cap rejections	23
Candidate states matching over four radii	2,516
Source calls: 281 targets at cap 64 / cap 16	2,881 / 3,152

Both public panels retain their recovery routes, with zero unresolved targets. The cap-16 panel preserves 260 later updates and 265 deferrals. Earlier representations remain as controls.

Costs and limits. Folding removes the balanced reciprocal matrix transform, but still stores four n-entry residues and performs phase gcds/modular powers. A direct V support can be larger than an unresolved W support. It removes quadratic-pair updates and reciprocal lifts from this path; no universal memory or timing advantage follows. All coefficient, source, initial-observation, order-check, allocation and final-verification work remains charged.

Evidence. validation01 passes with 88 source pins; inherited controls pass. No new compile/native/render failure or cloud/timing run. Producer assertion, independent signed verdict pending. Original 4,563-case cohort unchanged. Derivation and replay: DERIVATION.md, RESULTS.md, README.md.

Sources and next. Exact reciprocal/canonical/fused predecessor and official FLINT quotient conventions, flintlib.org/doc/nmod_poly/index.html. KB unavailable. Next investigate whether multiple observations yield a low-degree constraint before a degree-n initial support is constructed.